Lab Report Guidelines

Lab 2: Data Cleaning & Preprocessing for YouTube Text Mining

1. Cover Page

2. Objective

Briefly explain the purpose of this lab.

3. Tools and Libraries Used

Mention the Python version and all libraries used.

4. Dataset Description

Describe the datasets used in this lab:

- comments.txt: Raw YouTube comment data containing usernames, timestamps, and comment text.
- captions.vtt: Subtitle or caption data extracted from a YouTube video.

Mention any issues or irregularities noticed during the initial inspection (e.g., blank lines, “Reply” text, inconsistent formatting, or non-English text).

5. Methodology / Procedure 5.1 Profiling the Data

- Inspect raw .txt and .vtt files.
- Identify the structure and detect inconsistencies.

5.2 Parsing Comments and Captions

- Convert comments into a structured DataFrame with columns such as username, timestamp_text, and comment_text.
- Extract caption sentences using webvtt-py or manual parsing techniques.

5.3 Cleaning and Preprocessing Pipeline

Include short explanations or screenshots of your code for the following:

- Normalization (lowercasing, punctuation removal)
- Tokenization

- Stopword Removal
- Lemmatization or Stemming

5.4 Exporting Cleaned Data

- Save cleaned DataFrames as cleaned_comments.csv and cleaned_captions.csv.

6. Code Implementation

Provide the final, well-commented code used in this lab.
You may divide this section into subsections for clarity:

- Parsing Comments
- Parsing Captions
- Cleaning Functions
- Combined Cleaning Pipeline

7. Results and Discussion

Summarize the outputs and observations. Include screenshots or small tables if needed.

Include:

- Sample of cleaned tokens
- Number of valid comments/captions parsed
- Number of stopwords removed
- Any unexpected issues (e.g., missing lines, special characters, mixed languages)

Discuss:

- Differences between raw and cleaned data
- Challenges faced
- How preprocessing improved data quality and readability

8. Conclusion
Summarize your main findings and explain why this cleaning process is important for future labs.

9. References
List all materials, manuals, and external sources used.
Example:

- NLTK Documentation
- Pandas Documentation
- Python re Module Documentation
- webvtt-py Documentation

Submission Requirements:
Please upload the following files:

1. Lab Report (PDF format)
2. Python code file (.py or .ipynb)
3. Cleaned datasets (cleaned_comments.csv, cleaned_captions.csv)

Deadline: Submit before the assigned due date.